

Figure 1: The authentication login

Configure basic authentication for Apache:

```
# echo "test101" | htpasswd -i -c /etc/
httpd/.htpasswd admin
```

Now, add the following lines at the end of `/etc/httpd/conf/httpd.conf`:

```
<Directory "/var/www/html">
    AuthType Basic
    AuthName "Helm Private Repo"
    AuthUserFile /etc/httpd/.htpasswd
    Require valid-user
</Directory>
```

Restart `httpd` service to enable changes:

```
# systemctl start httpd
# systemctl enable httpd
# systemctl status httpd
```

Go to your browser and visit `https://localhost` to verify Web server status and basic authentication.

Create a Helm repository

We have the Web server up and running to host our Helm repository. Now, create a Helm repository inside the Apache Web root directory:

```
# mkdir /var/www/html/charts
# cd /var/www/html
# helm repo index charts/ --url https://
localhost/charts
```

Once done, you can visit `https://localhost/charts/index.yaml` and get the output shown in Figure 2.

To add a basic Helm Chart, download `https://github.com/Akasurde/shared_contents/raw/master/helm_chart/mychart-0.1.0.tgz` to the chart's directory.

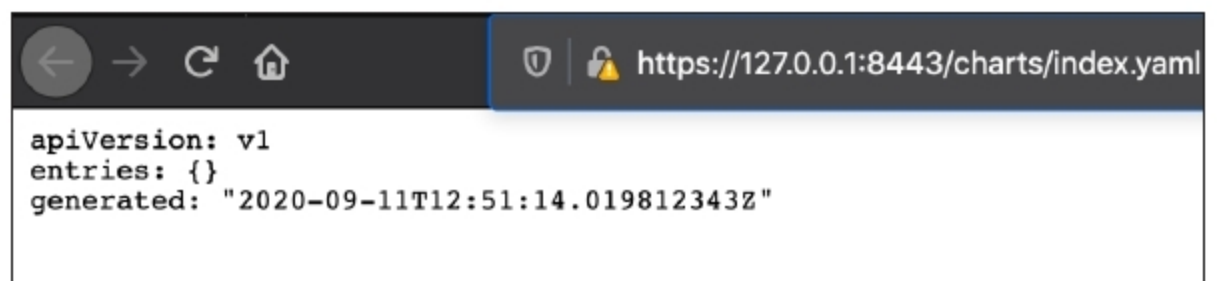


Figure 2: Chart output

```
# helm repo list
NAME      URL
stable    https://kubernetes-charts.storage.googleapis.com/
bitnami    https://charts.bitnami.com/bitnami
centos     http://localhost/charts
```

Figure 3: The list of repositories

```
# helm install mychart centos/mychart
NAME: mychart
LAST DEPLOYED: Fri Sep 11 19:28:22 2020
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/name=mychart,app.kubernetes.io/instance=mychart" -o jsonpath="{.items[0].metadata.name}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl port-forward $POD_NAME 8080:80
```

Figure 4: Installation confirmation

```
# kubectl -n test get pods
NAME                READY   STATUS    RESTARTS   AGE
mychart-5f789dc-6gsrx 1/1     Running   0           115s
```

Figure 5: Result of `get pods`

Add this newly created repository to the Kubernetes cluster:

```
# helm repo add centos https://
localhost/charts/ --insecure-skip-tls-
verify --username admin --password
admin
```

This will add the new repository `centos` in the Helm Chart repository. You can cross-verify by typing:

```
# helm repo list
```

Install Helm Chart

Install the Helm Chart from the newly added repository by using the following code:

```
# helm install centos/mychart mychart-
web
```

You can check your deployed application using:

```
# kubectl get pods
```

So, we have successfully created a private Helm repository and deployed a Helm Chart from this repository. **END** 🐧

Reference

Helm GitHub page - <https://github.com/helm/helm/>

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